

NAME

`_exit`, `_Exit` – terminate the calling process

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <unistd.h>
```

```
[[noreturn]] void _exit(int status);
```

```
#include <stdlib.h>
```

```
[[noreturn]] void _Exit(int status);
```

Feature Test Macro Requirements for glibc (see `feature_test_macros(7)`):

```
_Exit():
```

```
_ISOC99_SOURCE || _POSIX_C_SOURCE >= 200112L
```

DESCRIPTION

`_exit()` terminates the calling process "immediately". Any open file descriptors belonging to the process are closed. Any children of the process are inherited by `init(1)` (or by the nearest "subreaper" process as defined through the use of the `prctl(2)` `PR_SET_CHILD_SUBREAPER` operation). The process's parent is sent a `SIGCHLD` signal.

The value `status & 0xFF` is returned to the parent process as the process's exit status, and can be collected by the parent using one of the `wait(2)` family of calls.

The function `_Exit()` is equivalent to `_exit()`.

RETURN VALUE

These functions do not return.

STANDARDS

`_exit()` POSIX.1-2008.

`_Exit()` C11, POSIX.1-2008.

HISTORY

POSIX.1-2001, SVr4, 4.3BSD.

`_Exit()` was introduced by C99.

NOTES

For a discussion on the effects of an exit, the transmission of exit status, zombie processes, signals sent, and so on, see `exit(3)`.

The function `_exit()` is like `exit(3)`, but does not call any functions registered with `atexit(3)` or `on_exit(3)`. Open `stdio(3)` streams are not flushed. On the other hand, `_exit()` does close open file descriptors, and this may cause an unknown delay, waiting for pending output to finish. If the delay is undesired, it may be useful to call functions like `tcflush(3)` before calling `_exit()`. Whether any pending I/O is canceled, and which pending I/O may be canceled upon `_exit()`, is implementation-dependent.

C library/kernel differences

The text above in DESCRIPTION describes the traditional effect of `_exit()`, which is to terminate a process, and these are the semantics specified by POSIX.1 and implemented by the C library wrapper function. On modern systems, this means termination of all threads in the process.

By contrast with the C library wrapper function, the raw Linux `_exit()` system call terminates only the calling thread, and actions such as reparenting child processes or sending `SIGCHLD` to the parent process are performed only if this is the last thread in the thread group.

Up to glibc 2.3, the `_exit()` wrapper function invoked the kernel system call of the same name. Since glibc 2.3, the wrapper function invokes `exit_group(2)`, in order to terminate all of the threads in a process.

SEE ALSO

**execve(2), exit_group(2), fork(2), kill(2), wait(2), wait4(2), waitpid(2), atexit(3), exit(3), on_exit(3),
termios(3)**